

TYPED CONTROL, INFORMATION, AND SADDLE ENVELOPES FOR KINETIC COTANGENT PHASES

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ABSTRACT. Repeated noiseless observations are represented by an arbitrary-codimension current tower and an unnormalized evidence-bearing belief state. We prove the exact belief DPP, a two-saddle coarea/shell coefficient with separate numerator and denominator determinants, and exponential evidence containment. Reduction to kinetic variables is restricted to a strategy-reachable phase-chaotic class which retains phase weights and the finite correlation skeleton created by exact conditioning.

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1. ROUND-NINE RECONSTRUCTION: EVIDENCE-BEARING STRATIFIED BELIEFS AND A TWO-SADDLE FILTERING THEOREM

Exact observations are retained. Their posteriors are represented by unnormalized currents on a finite stratified observation tower together with the logarithmic evidence. This avoids a false uniform lower bound on the observation density. The exact belief state, including the phase weights and the finite correlation skeleton produced by conditioning, is Markov. A reduction to density/control coordinates is made only on a quantitatively

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reachable chaotic class and after the full belief dynamic programming theorem has been proved.

1.1. Admissible exact observation channels. Fix a finite observation schedule. An admissible channel on a chronological hard-sphere cell is a C^{r+3} map

$$O_k^u : E_\varepsilon \longrightarrow \mathbb{R}^{r_k}$$

chosen from a compact definable/Whitney-stratified family. On every stratum, O_k^u has constant rank; intersections of observation level sets with collision faces are clean and have uniformly bounded second fundamental form. Critical strata are included as separate lower-dimensional channels rather than declared part of the grazing set.

Let $\mathfrak{C}^{(c)}$ be the weighted normal-current space of codimension c . Boundary/collision tracing maps $\mathfrak{C}^{(c)}$ to $\mathfrak{C}^{(c+1)}$ at a clean intersection. Put

$$\mathfrak{B} = \bigoplus_{c=0}^{c_*} \mathfrak{C}^{(c)},$$

where c_* is the finite total observation/collision codimension over the fixed schedule.

Theorem 1.1 (Closed observation–collision current tower). *The B2 Green-graph propagator, every admissible coarea observation map, and specular collision tracing act continuously between the declared levels of $\mathfrak{B}$. For a prediction current G^- and an observed value y , the unnormalized posterior current is*

$$\mathcal{O}_{k,y}^u G^- = \delta_{O_k^u=y} G^-,$$

interpreted by the stratified coarea formula. It satisfies the exact Bayes identity and remains in $\mathfrak{B}$.

Proof. On one constant-rank stratum, choose normal coordinates for O_k^u and apply the coarea formula to the divergence-measure current. Clean intersection with a collision face makes the two normal traces commute up to the oriented intersection sign. Uniform Whitney estimates bound the resulting current norm. Sum over the finite stratification and use the B2 closed Green graph to pass to the completion. The coarea disintegration theorem gives the Bayes identity without dividing by the evidence. $\square$

Choose once and for all a smooth positive reference volume dy on each observation stratum. The evidence density and its logarithm are

$$Z_k^u(y; G^-) = \langle 1, \mathcal{O}_{k,y}^u G^- \rangle, \quad e_k = \begin{cases} \log Z_k^u(y; G^-), & Z_k^u > 0, \\ -\infty, & Z_k^u = 0. \end{cases}$$

The normalized posterior is stored only when $Z_k^u > 0$; the state always retains $(\mathcal{O}_{k,y}^u G^-, e_k)$, so small evidence does not enlarge its unnormalized current norm.

Lemma 1.2 (Reference covariance of evidence). *Replacing dy by $r(y)dy$, with r smooth and positive, changes e_k by $-\log r(y)$ and leaves the normalized posterior and every path likelihood ratio invariant. Thus the pair “unnormalized current plus evidence section” is an intrinsic density section of the observation line bundle.*

Proof. Both statements follow directly from the Radon–Nikodym transformation rule for the coarea marginal. In all ratios the same reference factor cancels. $\square$

1.2. The exact belief transition. Let Π_k be a probability measure on the augmented microscopic state or, equivalently on the moment-determinate reachable class, its complete current hierarchy in $\mathfrak{B}$. Given a control, first predict by the exact B2 law semigroup; then update by the observed current and its evidence.

Theorem 1.3 (Feller belief/evidence kernel and exact DPP). *On compact reachable current sublevels the map*

$$(\Pi_k, u, y) \mapsto \left(\frac{\mathcal{O}_{k,y}^u \Pi_k^-}{Z_k^u(y; \Pi_k^-)}, \log Z_k^u(y; \Pi_k^-) \right)$$

is continuous wherever $Z > 0$, and its unnormalized extension is Feller through $Z = 0$. The controlled sequence of complete beliefs is a Markov state under nonanticipative strategies. The finite-horizon risk-sensitive value satisfies the exact belief-state dynamic programming recursion.

Proof. Prediction is Feller by the B2 augmented law semigroup. On each Whitney stratum, coarea currents depend continuously on the state, control, and observation in the weighted weak topology; the finite frontier condition handles changes of stratum. Keeping the unnormalized current removes the singularity at $Z = 0$. Ionescu–Tulcea applied to the prediction and observation kernels constructs the strategy-dependent law. Conditioning at the next observation time gives the dynamic programming identity. Measurable selection follows from compact controls and the Feller kernel. $\square$

1.3. A two-saddle exact observation coefficient. Let H be the path/contact source accumulated before the observation and let $\zeta \in \mathbb{R}^{r_k}$ be the Fourier source for the observed variable. The joint full B2/B1 pressure is

$$Q_\varepsilon(H, \lambda, \zeta) = \mu_\varepsilon^{-1} \log \mathbb{E} e^{\mu_\varepsilon \langle H, \mathcal{X}_\varepsilon \rangle + \lambda C_\varepsilon + i \mu_\varepsilon \zeta \cdot \mathcal{O}_k^u}.$$

Admissibility includes a joint affine-span/constant-rank condition for the constraint and observation marks.

Theorem 1.4 (Joint coarea local coefficient). *For a regular observation value y and a compact source/control chart, the numerator and denominator of the conditional source expectation have separate exact finite saddles*

$$(\lambda_{H,y,\varepsilon}^N, \zeta_{H,y,\varepsilon}^N), \quad (\lambda_{y,\varepsilon}^D, \zeta_{y,\varepsilon}^D).$$

Their coarea densities satisfy uniform saddle expansions with their own Hessians and determinants. Consequently

$$\mathbb{E} \left[e^{\mu_\varepsilon \langle H, \mathcal{X}_\varepsilon \rangle} \mid O_k^u = y, \mathcal{S}_\varepsilon \right] = \exp \{ \mu_\varepsilon (\mathcal{Q}_N(H, y) - \mathcal{Q}_D(y)) + o(\mu_\varepsilon) \} \frac{d_N(H, y)}{d_D(y)} (1 + o(1)),$$

where the two rate terms and the two determinant coefficients are evaluated at their respective saddles. No common-saddle cancellation is assumed.

Proof. Insert the exact coarea delta through Fourier inversion in the observation variable and the B1 Fourier representation of the shell. The regenerative full-rank blocks are chosen for the combined constraint–observation mark, so the joint characteristic function has the central, minor-arc, and high-frequency bounds of B1 uniformly over the admissible observation family. Apply steepest descent separately to the source numerator and zero-source denominator. Their critical equations differ when $H \neq 0$, which produces the two saddles and determinant ratio. The stratified coarea Jacobian is included in each amplitude. $\square$

Corollary 1.5 (Evidence-tail large deviations). *For every M there is a compact reachable belief/evidence set such that the probability, uniformly over admissible strategies, of leaving it is at most $e^{-M\mu_\varepsilon}$. In particular observations with exponentially small evidence are retained with their correct risk-sensitive cost rather than excluded by a false deterministic lower bound.*

Proof. Theorem 1.4 gives the observation marginal rate and uniform prefactor on every stratum. Combine it with B2 exponential containment of the prediction current and take a finite cover of the compact control/stratum family. $\square$

1.4. Reachable phase-chaotic beliefs. An exact observation may create order-one correlations. The reduced state is therefore not just the one-particle mean. Let

$$\mathbf{b} = (f, \rho, \kappa)$$

contain the density f , the compact phase weight vector ρ , and a finite correlation skeleton κ consisting of all observation/collision strata which survive at order one. A belief is in the reachable chaotic class $\mathfrak{R}_T$ if its remaining connected correlations obey the B2 tree norm

$$\|h_j\| \leq C^j \mu_\varepsilon^{1-j}$$

after conditioning on (f, ρ, κ) .

Lemma 1.6 (Quenched propagation of the reachable cluster). *The prediction and admissible observation updates map $\mathfrak{R}_T$ into itself. Uniformly over strategies, all correlation coordinates outside (f, ρ, κ) change bounded dynamic rewards and transition pressures by $o(1)$.*

Proof. B2 source-uniform tree estimates propagate the connected hierarchy between observations. At an observation, differentiate the joint coarea pressure of Theorem 1.4; the finite set of order-one strata is placed in κ , while

each additional connected label pays the usual μ^{1-j} factor. Normal convergence makes the tail uniformly summable. Every bounded reward/source derivative against the tail is therefore $o(1)$. The statement is only for the strategy-reachable class, not for all singular beliefs sharing one mean. $\square$

Theorem 1.7 (Reduced kinetic game and statistical tangent). *The exact belief DPP of Theorem 1.3, restricted to $\mathfrak{R}_T$, converges locally uniformly to the DPP on the reduced state (f, ρ, κ) with transition Hamiltonians supplied by B4 and observation costs supplied by Theorem 1.4. At a regular identified phase, the finite experiment is LAN when centred at the exact finite score; with a prior positive at that phase, the posterior has the corresponding Bernstein-von Mises Gaussian limit. At coexistence the finite phase weights remain part of the limit state.*

Proof. Lemma 1.6 gives an exponentially contained, uniformly good approximation of the complete belief transition by the reduced transition. Apply dynamic programming backwards over the fixed observation schedule and use B4 comparison on every prediction interval. For LAN, Taylor-expand the joint source/observation pressure at its exact finite saddle; B3 gives the joint Gaussian score process and Theorem 1.4 gives the local likelihood ratio. Standard finite-dimensional Laplace expansion gives BvM on an identified chart. If several phases have the same leading cost, their separate evidence coefficients update ρ rather than being collapsed. $\square$

SCOPE AND VERIFICATION STATUS

This manuscript states positive theorem closure in the declared model-specific regular regime. Every cross-paper input is typed in the round-nine dependency ledger. Repository proof and build gates are internal checks and do not replace independent external mathematical review.

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